

PhishShield

Conversational Phishing Detection & Awareness Assistant Bot

Author: Shreyas Sandeep N

Date: 13/08/2026

Platform: Microsoft Copilot Studio (Microsoft Power Platform)

GitHub repository: [PhishShield](#)

Overview

PhishShield is a phishing detection chatbot developed using Microsoft Copilot Studio to help users identify potentially malicious emails and messages through a conversational interface. The bot analyzes user-provided content using pattern-based detection techniques and classifies it into High Risk, Medium Risk, or Safe categories based on the presence of common phishing indicators such as urgency, suspicious links, and requests for sensitive information.

In addition to risk classification, PhishShield provides pattern analysis, security recommendations, and phishing awareness tips to help users make informed decisions. The project was designed as a foundational cybersecurity solution with future scope for advanced capabilities such as AI-driven anomaly detection, URL intelligence, threat intelligence integration, and sandbox-based analysis.

Problem statement

Phishing remains one of the most common attack vectors used by cybercriminals to gain unauthorized access to systems, credentials, and sensitive information. While technology professionals are generally more aware of phishing techniques, they can still become victims in high-volume communication environments where messages are processed quickly. The risk is even greater for non-technical users, elderly individuals, and people who are new to digital platforms or the IT industry, as they may be less familiar with phishing indicators and social engineering tactics.

The impact of phishing extends beyond individual users. Organizations face financial losses, reputational damage, data breaches, and operational disruptions when malicious emails successfully bypass human judgment. With billions of malicious emails circulated globally each day, phishing continues to be a high-volume and highly effective attack method. Attackers frequently impersonate trusted organizations, financial institutions, and well-known brands to increase the likelihood of success.

PhishShield was conceptualized as an initial step towards addressing this challenge by providing users with a simple and accessible way to evaluate suspicious messages. The project aims to demonstrate how conversational AI can assist users in identifying common phishing patterns while laying the foundation for future enhancements, such

as automated phishing detection, anomaly analysis, and intelligent flagging of suspicious communications.

Objectives

1. Identify Suspicious Content

To analyze user-provided emails and messages for common phishing indicators such as suspicious links, urgency-based language, and requests for sensitive information.

2. Categorize Risk Levels

To classify analyzed content into appropriate risk categories (High Risk, Medium Risk, or Safe) based on the indicators detected.

3. Improve User Awareness

To provide users with recommendations and phishing awareness tips that help them make informed decisions when interacting with potentially suspicious communications.

4. Establish a Foundation for Future Enhancements

To create a scalable proof-of-concept that can be extended with advanced capabilities such as AI-driven anomaly detection, URL intelligence, automated phishing flagging, and threat intelligence integration.

Technologies used

Microsoft Copilot Studio

Used as the primary platform to design and develop the chatbot. It enabled the creation of conversational flows, user interactions, and phishing detection logic through a low-code interface.

Microsoft Power Platform

Provided the underlying ecosystem within which Copilot Studio operates, supporting bot development and workflow management.

Conversational AI

Used to deliver an interactive user experience, allowing users to analyze suspicious emails and messages through natural conversations.

Rule-Based Pattern Matching

Implemented to identify common phishing indicators such as urgency, suspicious links, and requests for sensitive information.

Conditional Logic & Formula Expressions

Used to process user input, evaluate phishing indicators, and classify content into High Risk, Medium Risk, or Safe categories.

Cybersecurity Awareness Concepts

Provided the foundation for detection logic by incorporating common phishing tactics and social engineering techniques into the analysis process.

System Design

The PhishShield system follows a layered conversational architecture that enables users to submit suspicious messages or emails and receive risk-based phishing assessments.

Input Layer

The user begins by selecting the type of content to analyze:

- Email
- Message

Once selected, the user submits the content for analysis through the chatbot interface.

Processing Layer

The submitted content is normalized and analyzed using rule-based pattern matching.

Key processing functions include:

- *Text()* – Converts user input into a string format.
- *Lower()* – Normalizes text to enable case-insensitive matching.
- *IsMatch()* – Identifies predefined phishing indicators.
- *Or()* – Combines multiple matching conditions.

The system evaluates the content against common phishing indicators such as:

- Suspicious links (http)
- Urgency-based language (urgent)
- Sensitive information requests (password)
- Financial references (bank)
- Action-oriented phrases (click)
- Verification requests (verify)

Decision Layer

Based on the detected indicators, the chatbot applies hierarchical classification rules:

1. High Risk
 - Strong phishing indicators detected.
 - Example: Links, credential requests.
2. Medium Risk
 - Suspicious behavioral patterns detected.
 - Example: Urgency or persuasive language.
3. Safe
 - No major phishing indicators detected.

Output Layer

The bot generates a structured response containing:

- Analysis Result
- Pattern Analysis
- Detected Indicators
- Security Recommendation

Users are then given the option to receive phishing awareness tips, making the solution both a detection and educational tool.

System Flow

This design provides a lightweight, explainable, and scalable foundation for phishing detection while allowing future integration of advanced capabilities such as anomaly detection, threat intelligence, URL reputation analysis, and sandbox-based inspection.

Key Logic & Expressions

Input Processing

Topic.CheckType

Pattern Matching Logic

Or(

IsMatch(Lower(Text(Topic.CheckType)), "http"),

IsMatch(Lower(Text(Topic.CheckType)), "urgent"),

IsMatch(Lower(Text(Topic.CheckType)), "password"),

```
IsMatch(Lower(Text(Topic.CheckType)), "bank"),  
IsMatch(Lower(Text(Topic.CheckType)), "click"),  
IsMatch(Lower(Text(Topic.CheckType)), "verify")  
)
```

This expression evaluates the presence of predefined phishing indicators and forms the primary detection mechanism of the chatbot.

Implementation

PhishShield implementation was divided into 3 key modules, each responsible for a specific stage of the phishing detection workflow.

Module 1: Content Input & Interaction Management

This module handles user interaction and acts as the entry point of the system. Users are first prompted to select the type of content they wish to analyze:

- Email
- Message

Once the selection is made, the bot requests the content to be analyzed and stores the user input for further processing. This module ensures a structured and user-friendly interaction flow while supporting multiple content types.

Module 2: Pattern Analysis Engine

The Pattern Analysis Engine forms the core processing layer of PhishShield. User-provided content is normalized using functions such as *Text()* and *Lower()* to ensure consistent evaluation. The system then performs rule-based pattern matching using *IsMatch()* against predefined phishing indicators, including:

- Suspicious links
- Urgency-related language
- Requests for sensitive information
- Verification requests
- Financial references
- Action-oriented instructions

This module simulates lightweight phishing analysis by identifying patterns commonly associated with phishing and social engineering attempts.

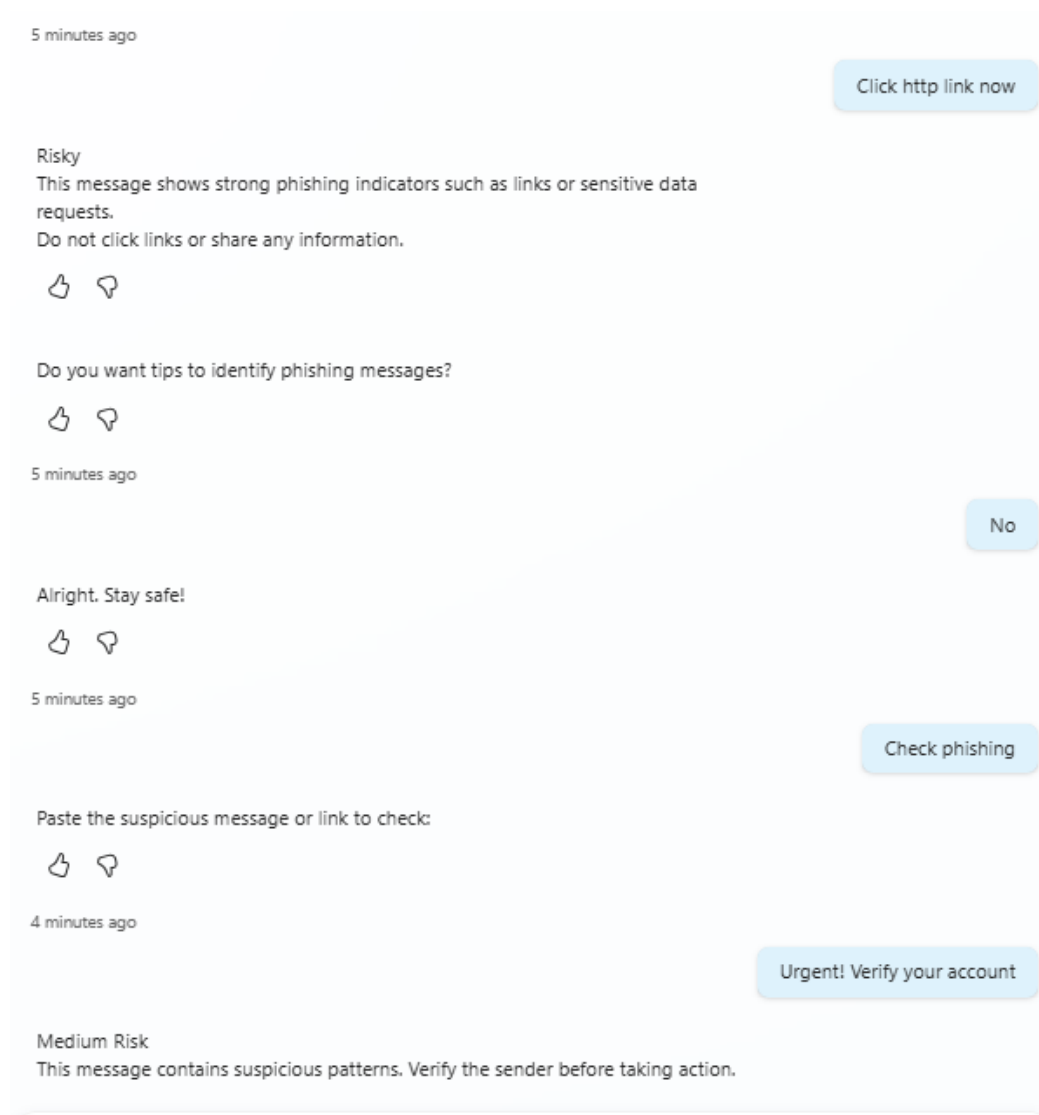
Module 3: Risk Classification & Guidance System

Based on the indicators identified by the Pattern Analysis Engine, the content is classified into one of three categories:

- High Risk
- Medium Risk
- Safe

The classification result is then presented to the user along with:

- Pattern analysis summary
- Detected indicators
- Security recommendations



Additionally, users are provided with an option to receive phishing awareness tips, enabling the system to function as both a detection assistant and a cybersecurity awareness tool.

Results

The developed PhishShield chatbot successfully demonstrated the ability to analyze user-provided emails and messages and classify them into appropriate risk categories based on predefined phishing indicators. The system was able to identify common phishing patterns such as urgency-based language, suspicious links, and requests for sensitive information, and subsequently categorize them as **High Risk**, **Medium Risk**, or **Safe**.

The implementation also successfully provided users with:

- Risk-based classification
- Pattern analysis summaries
- Security recommendations
- Optional phishing awareness tips

Testing across multiple scenarios showed consistent behavior and accurate routing through the chatbot's classification workflow. The addition of Email and Message analysis modes further improved the user experience by providing a more structured and realistic analysis process.

Despite its limited scope, the project achieved its primary objective of creating a simple, explainable, and user-friendly phishing awareness assistant while establishing a foundation for future cybersecurity-focused enhancements.

Challenges & Solutions

Several technical and design challenges were encountered.

1. Incorrect Risk Classification

Challenge:

The bot initially routed most inputs to the Safe category, even when suspicious indicators were present.

Solution:

The detection logic was refined by replacing the basic condition approach with formula-based pattern matching using functions such as *IsMatch()*. This improved the reliability of phishing indicator detection and classification.

2. Variable Handling and Input Processing

Challenge:

User input was not being consistently recognized during formula evaluation, leading to inaccurate matching and unexpected behavior.

Solution:

Proper variable referencing was implemented using *Topic.CheckType*, along with text normalization functions such as *Text()* and *Lower()* to ensure consistent processing of user input.

3. Limited Indicator Coverage

Challenge:

The initial version relied on a small set of phishing indicators, resulting in some phishing-like messages being classified incorrectly.

Solution:

Additional indicators were introduced and grouped into stronger and moderate categories, enabling more accurate risk classification and reducing false classifications.

4. Conversation Flow Stability**Challenge:**

During testing, certain conversation paths remained active or did not terminate correctly, resulting in inconsistent flow behavior.

Solution:

The chatbot flow was restructured, and topic completion logic was refined to ensure predictable execution and cleaner user interactions.

5. Platform Constraints**Challenge:**

Copilot Studio imposed certain limitations on operator behavior and content filtering, which occasionally affected implementation decisions.

Solution:

Alternative formula-based approaches and response refinements were adopted to work within platform constraints while preserving the intended functionality of the bot.

Future Improvements & Enhancements

Limited with its scope, the PhishShield can go through several enhancements to improve effectiveness, accuracy, and usability.

- **AI-Driven Detection**
Incorporate AI models to identify suspicious patterns and anomalies rather than relying solely on predefined indicators.
- **Advanced Link Analysis**
Examine URLs for suspicious structures, domain impersonation attempts, and potentially malicious patterns.
- **Threat Intelligence Integration**
Compare domains, IP addresses, and links against known threat databases to improve detection accuracy.
- **Automated Phishing Flagging**
Enable automatic identification and reporting of suspicious communications within organizational environments.
- **Sandbox-Based Analysis**
Evaluate links or attachments in a controlled environment to detect malicious behavior before user interaction.

Key Features

1. Email Analysis Mode
2. Message Analysis Mode
3. Pattern-Based Phishing Detection
4. High / Medium / Safe Risk Classification
5. Security Recommendations
6. Phishing Awareness Tips
7. Conversational User Experience

Conclusion

PhishShield demonstrates how a simple conversational AI solution can be leveraged to address a common cybersecurity challenge: phishing detection and user awareness. By combining rule-based pattern matching with a structured conversational workflow, the bot is able to analyze suspicious emails and messages, identify common phishing indicators, and classify content into appropriate risk levels.

Beyond detection, the project emphasizes usability and cybersecurity awareness by providing users with actionable recommendations and educational guidance. Through its implementation in Microsoft Copilot Studio, PhishShield highlights how low-code technologies can be used to rapidly develop practical security-focused solutions.

While the current implementation serves as a foundational proof-of-concept, it establishes a framework that can be expanded with advanced capabilities such as AI-driven analysis, threat intelligence integration, and automated phishing detection. Overall, PhishShield demonstrates the potential of conversational systems to support safer digital communication and improve cybersecurity awareness among users.